

Why Use $\log_{10}$ for Interpellet Intervals in Kernel Density Estimation?

When dealing with interpellet intervals (i.e., the time gaps between successive pellets taken by mice), these interval lengths can span multiple orders of magnitude and are often right-skewed. In other words, most intervals might be relatively short, but a small fraction can be very long, creating a heavily skewed distribution.

Reason for the Logarithmic Transformation

- **Reducing Skewness:** A logarithmic transformation (such as $\log_{10}$) compresses large values more than small values, often bringing the data closer to a symmetric (or at least less skewed) distribution. This can be advantageous for many statistical and visualization purposes.
- **Stabilizing Variance:** If the variance of the raw interpellet intervals increases with the mean, taking logs can help stabilize the variance. This improves the performance of certain statistical techniques.
- **Interpretability in Orders of Magnitude:** Using $\log_{10}$ specifically (as opposed to natural log) can provide a more direct interpretation in terms of “powers of ten.” This is sometimes chosen for convenience or for consistency with certain behavioral or physiological contexts.

KDE with Log-Transformed Data

The kernel density estimate (KDE) formula is typically:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right).$$

When applying KDE to interpellet intervals, one might replace each raw interval x_i with $\log_{10}(x_i)$ before computing the density. The resulting KDE would describe the density of $\log_{10}$ -transformed values rather than the raw intervals themselves.

Practical Considerations

1. **Choice of Kernel and Bandwidth:** After transforming the data (e.g., taking logs), you still need to select an appropriate kernel function K and bandwidth h . The transformed data might be better-behaved, allowing for simpler bandwidth-selection methods.
2. **Back-Transformation:** If you want to interpret or visualize the density on the original time scale, you can apply an inverse transformation (exponentiation in base 10). Careful with any density or probability-axis adjustments needed when back-transforming.

3. **Zero or Negative Values:** Logarithms are undefined for non-positive values. Hence, you must ensure all interpellet intervals are strictly positive and handle any zeros or extremely small intervals carefully (e.g., by adding a small constant before taking the log).

Summary

Using $\log_{10}$ for interpellet intervals in the KDE formula is primarily a way to manage the highly skewed distribution of interval lengths. By compressing the scale of large values, the data often become more amenable to kernel smoothing techniques, and the resulting density estimates can be more informative and visually interpretable.